PROBLEM 2.2. The Viterbi Algorithm

a) Implement the basic Viterbi algorithm using `jax.lax.scan` and run it on data generated by the provided dataset generator. Try it out with chain lengths of 10, 20 and 100. What do you observe?

b) Reformulate and implement the algorithm using log likelihoods(LL). Rerun it with the chain lengths from Task a).

c) Extend the program such that it produces the $H = 3$ best hypotheses of the underlying state sequence and their likelihoods. Implementing for an arbitrary H will help you with the project.